

Holger Pirk

Follow up 1 - 14/4/2026

Prepared by
Session date

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Holger Pirk – Follow Up Consultation Notes

Date: 14 April 2026

Consultation Type: Follow-up (45 minutes)

Practitioner: Jenaed Ruddock RD

1. Current Status Overview

Performance & Training Context

Holger recently completed the Boston Half Marathon (Lincolnshire) in challenging wind conditions and achieved a **personal best of 1:28:50**.

- Performance indicates **clear improvement in fitness**
- However, perceived potential (~1:25) suggests **performance is not yet fully realised**

Clinical Interpretation:

This pattern is commonly seen in endurance athletes where:

- Fitness improves
 - But **energy availability limits performance expression**
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Training Load

- Weekly mileage: **70–90 km/week**
- Daily additional bodyweight training

Clinical Context:

This represents a **high chronic training load**, requiring:

- High total energy intake
 - Consistent carbohydrate availability
 - Structured recovery nutrition
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Body Composition & Weight Trends

- Current weight: **79.7 kg**
- Previous: ~78–78.5 kg
- **~1 kg increase over 14 days**

Interpretation:

- This is a **positive and expected response**
- Early weight gain reflects:
 - glycogen restoration
 - increased fluid retention
 - improved energy availability

Key Target:

- **~400g/week increase over 4 weeks**

This rate is:

- Sufficient to restore physiology
- Without excessive fat gain

2. Energy Intake Assessment

Current Intake

- **3400–3600 kcal/day (average)**

Interpretation:

- Likely:
 - Adequate for **easy days**
 - **Slightly insufficient for quality and long run days**

Important Concept:

Average intake can be misleading

→ If intake is too low on key days, the **weekly average still hides deficits**

Energy Availability (Key Concept)

Holger's presentation reflects **low energy availability (LEA)**:

This occurs when:

| Energy intake is insufficient relative to training load and physiological demands

Why this matters:

Low energy availability affects:

- Hormonal function (↓ T3)
- Recovery capacity
- Muscle repair
- Glycogen replenishment
- Nervous system function

3. Blood Markers – Detailed Clinical Interpretation

Thyroid Function (Most Important Marker)

- **Free T3: LOW (2.7 pmol/L)**
- TSH: Normal
- T4: Normal

Explanation:

- T3 is the **active thyroid hormone**
- It reflects **metabolic activity and energy availability**

When energy is low:

- The body reduces T3 to:
 - conserve energy
 - slow metabolism

Clinical Meaning:

This is a **functional adaptation**, not disease

→ The body is saying:

| “There is not enough fuel available — conserve energy”

Expected recovery:

- Requires **consistent adequate carbohydrate and energy intake**
- Typically improves over **3–6 months**, not immediately

Vitamin D Deficiency

- Level: **22 nmol/L (deficient)**

Explanation:

Vitamin D plays a role in:

- muscle function
- immune health
- inflammation regulation
- recovery

Current Plan:

- 50,000 IU weekly for 6–8 weeks
- Then maintenance: ~2000 IU/day

Clinical Note:

Endurance athletes often require **higher levels** due to:

- increased turnover
- higher physiological stress

Haematology (Red Blood Cells)

- Low Hb / RBC / HCT
- Ferritin normal

Explanation:

This is **not iron deficiency**

Instead:

- Plasma volume increases with endurance training
- This dilutes red blood cells → “sports anaemia”

Additional factor:

- **Foot-strike haemolysis**
 - red blood cells break down due to impact

Clinical Meaning:

- Normal adaptation to high training
- Does **not require intervention**

Urea & Bicarbonate

- **High urea**
- **Low bicarbonate**

Explanation:

This reflects:

- Increased **protein breakdown**
- Body using **protein as fuel**

Why this happens:

When carbohydrates are insufficient:

- Body increases:
 - gluconeogenesis (making glucose from protein)
 - muscle breakdown

Clinical Meaning:

| This is a **catabolic state driven by under-fuelling**

Liver Enzymes (AST)

- Mild elevation

Explanation:

- Likely from:
 - muscle breakdown
 - training stress

Clinical Meaning:

- Common in endurance athletes
- Not clinically concerning

4. Nutrition & Fuelling Behaviour Analysis

Pre-Run Nutrition

Current:

- Porridge (~800 kcal, with peanut flour)
- Timing: ~2 hours pre-run

Positive:

- Good structure
- Appropriate for sustained energy

Limitation:

- Does not address:
 - **early-run glucose availability**
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Physiological concept:

First 20–30 mins of a run:

- body relies on **circulating glucose**
 - if unavailable → increases:
 - heart rate
 - perceived effort
 - muscle breakdown
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Recommendation:

Add:

- **15–30g fast carbohydrates 15–20 mins pre-run**
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During-Run Fuelling

Current:

- Inconsistent
 - Often not fuelling shorter runs
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Problem:

Even runs >45 mins:

- can become **catabolic without fuel**
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Mechanism:

Without carbohydrate intake:

- glycogen is used quickly
 - body shifts to:
 - protein breakdown
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Recommendation:

- All runs >45 mins:
→ **30–60g carbs/hour**

Post-Run Nutrition

Current:

- Long runs: ~45–50g carbs + 30g protein
- Easy runs: inconsistent

Key Issue:

Not quantity — **consistency**

Important concept:

The body responds to:

| **predictability of fuel availability**

Recommendation:

- Every run:
→ **something within 30 mins**

Even if small:

- yoghurt + fruit
- milk + cereal
- smoothie

5. Key Clinical Themes

1. Transition from Catabolic → Anabolic State

Currently:

- body still partially in **energy conservation mode**

Goal:

- shift toward:
 - recovery
 - adaptation
 - performance

2. Carbohydrate Availability is the Primary Lever

Not protein

Carbohydrates:

- regulate T3
 - support glycogen
 - reduce muscle breakdown
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3. Consistency Over Perfection

More important than:

- exact macros
 - perfect meals
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6. Core Recommendations

1. Fuel ALL runs >45 minutes

- 30–60g carbs/hour
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2. Post-run nutrition within 30 mins

- Focus on:
 - consistency
 - accessibility
-

3. Pre-run quick carbohydrate

- 15–30g 15–20 mins pre-run
-

4. Increase energy intake on key days

- +300–500 kcal
 - Mostly carbohydrates
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5. Hydration strategy

- Add structured sodium + fluid intake
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6. Weight monitoring

- Target:
 - ~400g/week
 - Focus on trend
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Progress:

- Positive trajectory
- Early improvements visible

Limiting factor:

| Carbohydrate and total energy availability

Strategic focus:

- Stabilise intake
- Improve fuelling consistency
- Support physiological recovery